

Optimal Bidding of Li-ion BESS in Regulation Markets Considering Capacity Degradation

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Abstract—The grid deployment of lithium-ion (Li-ion) battery energy storage systems (BESS) has grown fast during recent years. One emerging grid application is using Li-ion BESS in frequency regulation markets for tracking fast-moving frequency regulation signals, such as PJM's RegD signal. However, Li-ion batteries may encounter fast capacity degradation when tracking these signals. To effectively incorporate the cycling-induced degradation cost into the optimal bidding in regulation markets, we develop a statistical feature-based method for capacity degradation estimation. The fractional absolute moment (FAM) of the state-of-charge (SoC) profile is selected as the input feature of a linear regression model that produces an estimate of the capacity degradation as the model's output. Subsequently, the optimal bidding problem is formulated as a convex optimization model to maximize the daily profit of Li-ion BESS in a regulation market. Simulation results show that the proposed model provides, for frequency regulation, cost-effective capacity bidding and 5-min economic basepoints which achieve a tradeoff between regulation credit and battery degradation cost.

I. INTRODUCTION

Lithium-ion (Li-ion) battery energy storage systems (BESS) have good performance characteristics, such as high energy density, good cycling stability, and low self-discharge, and have become a popular choice for a number of grid applications. One emerging grid application is the participation in pay-for-performance frequency regulation markets. Several recent studies in the literature investigated the optimal participation of Li-ion BESS in regulation markets. Chen et al. [1] proposed a co-optimization model for frequency regulation and energy arbitrage based on multi-scale dynamic programming, where the capacity bidding was assumed to be fixed. Their study was focused on the real-time control throughout an operation day. Although the revenue from frequency regulation appeared to be attractive, the study in [1] did not consider the fast capacity degradation that Li-ion BESS may experience while tracking fast-moving frequency regulation signals.

Researchers recently began to take into account degradation when optimizing BESS participation in regulation markets. Shi et al. [2] developed a joint optimization framework for peak shaving and frequency regulation, in which the battery capacity degradation is modeled to be a linear function of the throughput charge independent on the cycle depth. However,

many studies on Li-ion battery aging reveal that the cycle depth has a significant impact on the capacity degradation [3]–[6]. Xu et al. [7] used a polynomial stress function of the cycle depth to capture the cycling-induced aging and combined this stress function with the rainflow counting (RFC) method to estimate battery capacity degradation. However, RFC is difficult to be incorporated into battery optimization models due to the lack of an analytical formulation with respect to battery operation strategies [8]. As an effort to address this difficulty, Shi et al. [9] proved that the RFC-based capacity degradation estimation model is a convex function of battery output power provided that the embedded stress function of the cycle depth is convex. Despite the certificated convexity, the incorporation of RFC renders a battery optimization model non-differentiable [9]. A subgradient algorithm was used to solve the non-differentiable optimization problem, which took a relatively long computing time because the subgradient algorithm is often much slower than interior-point methods [10]. Also, such an incorporation of RFC in [9] necessitates the use of frequency regulation signals at every time point, which makes the optimization model unsolvable in practice due to the fact that frequency regulation signals are not known in advance [7].

In this study, we first propose a statistical feature-based capacity degradation estimation model that is convex and differentiable. We then formulate a convex optimization model for the Li-ion BESS participating in regulation markets, the solution of which yields the optimal regulation capacity bidding and economic basepoints.

II. MODEL FORMULATION

A. Regulation Market of PJM

We consider the PJM's pay-for-performance regulation market in our study. In this regulation market, the total price of a regulation offer is composed of the regulation capacity portion and the performance portion that reflects the regulation movement. Mileage is used to represent the movement of a specific type of regulation signal. The market-clearing engine ranks all potential regulation resources in the ascending order of their merit order prices indicating the regulation offer cost. The PJM's regulation market is cleared by selecting the offered

resources with the lowest merit order prices to satisfy the regulation requirements (i.e., 525 effective MW during non-ramp hours and 800 effective MW during ramp hours [11]). The highest merit order price among the selected resources becomes the regulation market clearing price (RMCP). The final regulation market capacity clearing price (RMCCP) is derived from RMCP by subtracting the regulation market performance clearing price (RMPCP).

The fast-moving RegD signal is updated and then sent to faster, more dynamic resources (e.g., BESS) every two seconds, while the traditional RegA signal is used to dispatch slower and sustained-output resources, such as steam and combustion units. PJM monitors the responses of the resources being offered and calculates their performance scores at the end of each regulation hour. A resource's performance score must be greater than or equal to 75% in order for the resource to be eligible for providing regulation services.

PJM calculates the RMCCP credit for a regulation resource, R_c , by multiplying RMCCP, regulation capacity C_f , and the performance score γ_p , given by

$$R_c = \text{RMCCP} \cdot C_f \cdot \gamma_p \quad (1)$$

The RMPCP credit for a regulation resource is calculated by multiplying RMPCP, regulation capacity C_f , the mileage ratio of the RegD signal (for Li-ion BESS) over the traditional regulation signal RegA, and the resource's performance score γ_p . This credit takes the following form:

$$R_p = \text{RMPCP} \cdot C_f \cdot \frac{M_{\text{RegD}}}{M_{\text{RegA}}} \cdot \gamma_p \quad (2)$$

The total regulation credit is the sum of capacity and performance components, expressed as

$$R_f = R_c + R_p \quad (3)$$

B. Statistical Feature-based Capacity Degradation Estimation

The RFC method, used in conjunction with an empirical cycle-depth stress function, is a standard approach to estimating the battery capacity degradation in grid applications. However, the RFC method renders the degradation cost function non-differentiable. The use of such a degradation cost function in optimal frequency regulation thus results in a heavy computational burden. For example, it took 2,522 seconds to solve the optimal frequency regulation within a 24-hour horizon by using the subgradient algorithm [9].

The RFC method is originally applied in fatigue damage estimation, where several alternatives to the RFC method include the spectral method, stochastic method, and hysteresis operator [12]. Fatigue can be interpreted as the weakening or breakdown of a material undergoing stresses, in particular cyclic stresses [12]. The spectral method aims to approximate the rainflow density based on the spectral moments of the input signal. The stochastic method models fatigue as a stochastic process, which is suitable for time-varying random loads. The hysteresis operator method is used to calculate a fatigue life parameter under random loading, namely the dissipated energy.

For a BESS participating in the regulation market, the SoC will be the weighted integral of the RegD signal with random nature. We try to utilize the statistical information of the SoC to estimate the capacity degradation. For ensuring our estimation results are close to those by the RFC method, we select as the input feature the fractional absolute moment (FAM) of the SoC with the same order as the empirical cycle-depth stress function used in the RFC method and use this feature to estimate the daily capacity degradation. The proposed degradation estimation model can be expressed as

$$C^{\text{deg}} = \beta \sum_{h=1}^{24} \mathbb{E} |\text{SoC}_h - \mathbb{E}[\text{SoC}_h]|^\alpha \quad (4)$$

where α is the order of the empirical cycle-depth stress function, h is the index of hour, and the coefficient β can be identified by using the least squares regression.

C. Optimal Bidding Model for Frequency Regulation

Our goal is to maximize the daily profit of a Li-ion BESS participating in the frequency regulation market. The optimization variables are the hourly capacity bidding and 5-min economic basepoints for frequency regulation. Based on the policies in the PJM's frequency regulation and real-time energy markets, the daily revenue of the BESS is calculated as

$$R = \sum_{h=1}^{24} C_h^f (\text{RMCCP}_h + \text{RMPCP}_h \cdot \gamma_m) + \sum_{l=1}^{288} \frac{1}{12} c_l^r (p_{h(l)}^b + C_{h(l)}^f \bar{r}_l) \quad (5)$$

The first term in the sum represents the daily regulation credit, where C_h^f is the capacity bidding and γ_m is the mean RegD-to-RegA mileage ratio. The second term in the sum represents the daily revenue from the real-time energy market, where c_l^r is the locational marginal price (LMP), $\bar{r}_l$ is the mean value of the RegD signal in the l th 5-min interval, $p_{h(l)}^b$ is the economic basepoint for regulation. Since the revenue from the energy market is much less than the revenue from the regulation market, the economic basepoints are intended to compensate the mean energy consumption of the BESS when tracking the RegD signal. This energy consumption includes energy losses due to efficiencies less than 100%.

The daily degradation cost of the revenue is given by the product of the daily capacity degradation and the battery pack replacement cost.

$$C_d = c^b C_0 C^{\text{deg}} \quad (6)$$

where c^b is the unit cost of the battery pack per kWh and C_0 is the initial capacity of the BESS in kWh. Next, the optimization model is formulated to achieve an optimal balance between the daily regulation credit and degradation cost

$$\max_{C^f, \mathbf{p}^b} R_d - C_d \quad (7)$$

subject to output power limits (8-12), SoC constraints (13-14), and a minimum discharge time of 15 min at the bidding

regulation capacity (15)

$$p_h^b = \sum_{k \in K_h} C_h^f [\max(-r_k, 0) e^c - \max(r_k, 0) / e^d] / |K_h|, \forall h \quad (8)$$

$$-P_0 \leq p_h^b \leq P_0, \forall h \quad (9)$$

$$C_h^f \leq P_0, \forall h \quad (10)$$

$$p_h^b + C_h^f \leq P_0, \forall h \quad (11)$$

$$p_h^b - C_h^f \geq -P_0, \forall h \quad (12)$$

$$\text{SoC}_l = \text{SoC}_{l-1} - C_{h(l)}^f \left[\sum_{k \in K^l} \max(r_k + p_{h(l)}^b / C_h^f, 0) / e^d + \sum_{k \in K^l} \min(r_k + p_{h(l)}^b / C_h^f, 0) e^c \right] / (1800 C_r), \forall l \quad (13)$$

$$0.1 \leq \text{SoC}_l \leq 0.9, \forall l \quad (14)$$

$$C_h^f \leq 4 \text{SoC}_{l(h)} C_r, \forall h \quad (15)$$

where r_k is the RegD signal at the k th 2-s time point; K_h is the set of 2-s time points in the h th hour; P_0 is the nominal power of the BESS; e^c and e^d are the charge and discharge efficiencies; SoC_l is the SoC at the l th 5-min time point; C_r is the remaining capacity of the battery. The degradation cost is always a convex function for any real order of α , and all other functions involved in the objective function and constraints are linear. Therefore, the optimal regulation capacity bidding problem is formulated as a convex optimization model that can be readily solved by mature solvers.

III. CASE STUDY

We consider a 1MW/0.5MWh Li-ion BESS participating in the PJM's regulation market. The charge and discharge efficiencies are both set as 95%, and the battery pack replacement cost is set as \$209/kWh [13]. We use the 5-min real-time LMP and regulation market dataset from April 1, 2018 to December 28, 2018, available on the PJM website [14], [15].

First, we identify the unknown coefficient β of our FAM-based capacity degradation estimation model. The BESS is simulated to operate for 2,000 days by using the resampling technique. The coefficient β is estimated based on the least squares regression, as shown in Fig. 1. Then, we solve the convex programming problem defined in (7) and (8) by using CVX, a Matlab-based modeling system for disciplined convex programming. The computing time is on average 2.15 s, which is significantly shorter than the computing time of the method using RFC [9]. Figure 2 shows the optimization results on a randomly chosen day. The optimal daily profit is \$1,112, where the revenue from the frequency market is \$1,420, the energy charge paid to the real-time energy market is \$21, and the degradation cost is \$287.

The initial daily capacity degradation is estimated to be 0.275%, which means the battery lifetime is approximately 363 days if this degradation rate does not change over time.

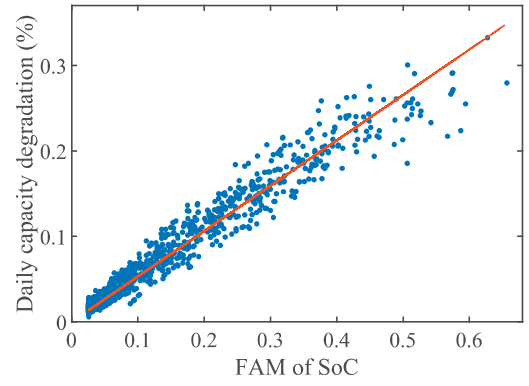


Fig. 1. Capacity degradation estimation of FAM-based method and RFC

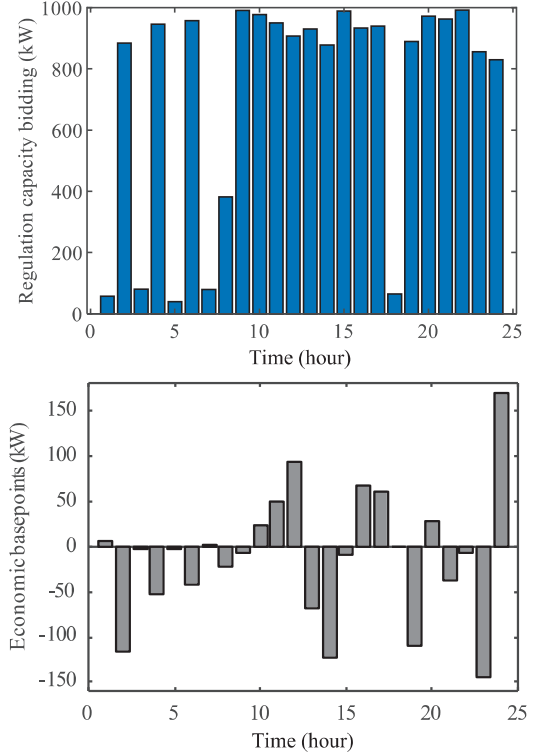


Fig. 2. Optimal bidding of regulation capacity and economic basepoints

However, as shown in Fig. 3, the degradation rate decreases as the regulation bidding of the BESS is optimized day-by-day until the remaining capacity reaches 100 kWh that is the minimum capacity to be eligible to participate in the PJM's regular market. The results show that the lifetime of the battery pack is close to 1,000 days and the life-span profit is \$70,332.

Table I presents a comparison between our method and two other methods of optimizing the participation of BESS in regulation markets. Our method considers that BESS perfectly track the regulation signal in the operation hour with offered regulation capacity, while the methods proposed in [7] and [9] consider that BESS strategically operate by manipulating the tracking performance.

TABLE I
COMPARISON OF THREE CONTEMPORARY METHODS FOR REGULATION CAPACITY BIDDING AND OPERATION.

Method	Focus	Degradation model	Computation time	Market role
Our method	Optimal capacity bidding with fully tracking operation	FAM	2.15 s	Price taker
Threshold-based method [7]	Optimal operation and segment capacity bidding	RFC	Real-time	Price maker
Convex RFC-based method [9]	Optimal operation with fixed capacity bidding	RFC	2,522 s	Price taker

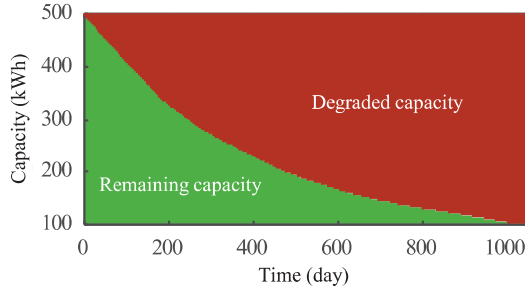


Fig. 3. Capacity degradation over operation days

IV. CONCLUSION AND FUTURE WORK

We propose a statistical feature-based model to estimate the capacity degradation of Li-ion BESS participating in regulation markets. The FAM of the SoC profile is selected as an input to estimate the capacity degradation. Simulation results show that this input feature has a high linear correlation with the capacity degradation (i.e., the linear correlation coefficient equal to 0.98). We formulate a convex optimization model incorporating the capacity degradation to determine the optimal bidding of regulation capacity and economic basepoints. Simulation results demonstrate that our model can achieve the optimal daily profit with a significant shorter computing time than the existing ones incorporating the RFC-based degradation estimation model. Our future work will focus on using experimental data of Li-ion battery aging to verify and refine the proposed statistical feature-based degradation estimation model.

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